

Short Term Interest Rate Innovations Across Monetary Regimes

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Draft

Abstract

This paper revisits the asset-pricing evidence in Maio and Santa-Clara (2017), who study short-term interest-rate innovations and stock-market anomalies. The contribution is a replication-first extension: the published design is audited before any temporal, regime, or high-frequency interpretation is allowed to enter the evidence set. The current draft reports a data-limited non-reproduction status rather than empirical magnitudes. The article and publisher supplement are verified through local hashes, the table targets are registered, and the estimation and validation code is tested, but source-compatible factor, portfolio, first-pass, cross-sectional, and extension artifacts are not yet available. The paper therefore identifies the claims that can be tested, the inputs that still block those tests, and the conditions under which a later empirical version may classify evidence as confirmation, narrowing, reinterpretation, generalisation, contradiction, or missing-input non-reproduction.

1 Introduction

Short-term interest rates are natural candidates for state variables in intertemporal asset-pricing models. A shock to the investment opportunity set can change discount rates, hedging demands, and the relative performance of portfolios whose cash-flow profiles or balance-sheet exposures differ. The empirical challenge is that a residual from an autoregressive model for a short rate is not automatically an economically narrow object. It may summarize monetary-policy surprises, central-bank information, macroeconomic news, data revisions, and sample-specific persistence. A replication that treats this residual as a priced factor must therefore separate the mechanical construction of the innovation from the economic interpretation attached to it.

Maio and Santa-Clara study this problem in the setting of stock-market anomalies. This paper starts from their published article and asks a narrower but more auditable question: when the baseline design is rebuilt from frozen sources, does the short-rate innovation remain a stable priced factor once evidence is separated by replication status, sample vintage, monetary regime, and high-frequency shock component? The paper is deliberately sequential. A table cannot be called reproduced until it is matched to a source-compatible input and an auditable generated statistic. An extension cannot revise the baseline verdict. A causal interpretation cannot be imported into the AR residual.

The current version is not an empirical-results paper. It is a working paper at the evidence-freeze boundary. The repository has a complete publication-document evidence pack and a tested pipeline for factor construction, portfolio assembly, time-series estimation, cross-sectional pricing, robustness diagnostics, temporal extension, monetary-regime analysis, shock decomposition, out-of-sample falsification, manuscript checks, and release checks. What it does not yet have is the source-compatible empirical panel required to populate the tables. The result is a data-limited non-reproduction draft: the design is executable and reviewable, while the numerical claims remain blocked.

2 Contribution

The first contribution is a table-level replication protocol. Rather than ask whether the paper is broadly reproducible, the repository freezes individual targets and assigns each one a status label. The allowed labels are reproduced, approximately reproduced, missing-input, contradicted, and not-attempted. This distinction matters because inaccessible author portfolios, ambiguous public archive versions, and unpublished construction details are not the same as statistical disagreement.

The second contribution is an extension design that preserves the baseline vintage. The post-publication analysis is not allowed to overwrite the original sample with revised data and then call the combined file a replication. Extension evidence is recorded as new evidence, with its own sample window, source vintage, asset universe, and uncertainty method.

The third contribution is an interpretation boundary. The short-rate residual in the baseline is a rate innovation. A high-frequency decomposition can later split announcement surprises into narrower components, but that identification exercise belongs to a separate section and cannot be retrofitted into the baseline language.

3 Model

The baseline model is the two-factor beta representation used for the replication target. For portfolio i , excess return R_i^e , market factor m , and short-rate innovation u_r , the time-series regression is

$$R_{i,t}^e = a_i + \beta_{i,m}m_t + \beta_{i,r}u_{r,t} + \varepsilon_{i,t}.$$

The cross-sectional pricing equation is

$$E[R_i^e] = \lambda_0 + \beta_{i,m}\lambda_m + \beta_{i,r}\lambda_r + \alpha_i.$$

The empirical question is whether the estimated $\beta_{i,r}$ exposures help explain average excess returns after accounting for market exposure and the registered comparator models.

The strict baseline sample is configured from 1972-01 through 2013-12. For any baseline-results paragraph, the sample is the configured monthly baseline, the model is the market-plus-short-rate pricing system, the first-pass estimator is ordinary least squares with an intercept, the second-pass estimators are the registered cross-sectional procedures, the test assets are the registered anomaly portfolios, uncertainty uses the configured Newey-West and Shanken-style adjustments where applicable, and economic magnitude is not reported until generated cells exist.

4 Evidence

The publication evidence is frozen at the document level. The final article PDF and publisher supplement are available only as local private research copies, and the committed repository records their metadata and hashes rather than the files themselves. The remaining evidence is source-level and artifact-level: source definitions, data-access status, table targets, generated audits, and generated reports.

This table is not a results table. It is a map from each claim-bearing object to the artifact that must support it. Its purpose is to prevent the paper from turning inaccessible inputs into an accusation or turning partial reconstruction into exact replication.

Table 1: Current Evidence Register

Evidence object		Machine-readable source	Current status
Publication documents	docu-	artifacts/evidence/article_manifest.json	Hashes recorded, files local-only
Source access		research/data_access_matrix.csv	Several exact definitions pending
Table targets		research/table_target_manifest.csv	Baseline and appendix targets frozen
Baseline audit		artifacts/audit/table_replication.csv	Missing-input status only
Release gate		artifacts/release/release_gate.json	Source-only release permitted

5 Replication Design

The replication design has three layers. The first layer freezes the bibliographic target and the table-level acceptance criteria. The second layer records every source needed to build the baseline panel, including whether the exact definition has been verified and whether redistribution is permitted. The third layer runs the estimators and assigns table-level labels only after generated artifacts can be compared to the registered targets.

The factor reconstruction starts from a monthly short-rate series and estimates an autoregressive innovation with an intercept. The portfolio reconstruction uses the configured anomaly portfolio sets and preserves the distinction between public French-library data, author-provided portfolios, and WRDS-style reconstructions. The estimator reconstruction uses time-series regressions for betas and cross-sectional pricing regressions for risk prices and pricing errors. Numerical tolerances are registered before results are interpreted.

6 Baseline Replication Status

The baseline audit currently classifies the frozen targets as missing-input rows. For this status paragraph, the sample is the strict monthly baseline, the model is the registered baseline and comparator system, the estimator is the table-level audit over generated cells, the test assets are the target anomaly portfolios, uncertainty is unavailable because generated standard errors do not exist, and economic magnitude is unavailable because no replicated table cell has been produced.

This status has a narrow meaning. It says that the current repository cannot yet produce the empirical tables from source-compatible inputs. It does not say that the published article is contradicted, and it does not license substituting a nearby public file while calling the output exact replication. Once the required inputs are registered, the same audit file will be regenerated and each target will move to a reproduced, approximately reproduced, contradicted, or still missing-input label.

7 Robustness And Weak-Factor Design

The robustness layer is designed to stop a high fitted cross section from being overread. The pipeline records beta dispersion, beta-matrix rank, singular values, factor spanning, influence diagnostics,

leave-one-family pricing systems, and familywise adjustments. The baseline factor must be strong enough, and variation in the estimated loadings must be informative enough, before a risk price can carry economic interpretation.

For robustness paragraphs, the sample is the generated strict baseline, the model is the registered robustness family, the estimator is diagnostic comparison over the baseline pricing artifacts, the test assets are the same assets used for the replication audit, uncertainty is handled through weak-factor and familywise diagnostics, and economic magnitude is blocked until baseline estimates exist.

8 Temporal Extension

The temporal extension asks whether a relationship estimated in the publication sample survives in later data. The baseline window remains frozen through 2013-12, and the extension window starts in 2014-01. Those dates are not results. They are design constraints that prevent the extension from revising the evidence used to classify the baseline replication. The extension also records revised-history audits, compatible asset universes, and frozen-parameter pricing comparisons.

For temporal-extension paragraphs, the sample is the frozen baseline plus the configured post-publication window, the model is the frozen-parameter and refit extension design, the estimator is the predeclared temporal comparison, the test assets are source-compatible baseline assets, uncertainty comes from the temporal diagnostics, and economic magnitude is blocked until extension panels exist.

9 Monetary Regimes

The regime analysis is an extension, not a baseline replication device. Regime boundaries are declared before any pricing result is inspected, and the design includes boundary-shift checks, pooled interactions, split-sample eligibility rules, and structural-break diagnostics. A regime finding may narrow or reinterpret the baseline evidence, but it cannot be used to relabel a published table target.

For regime paragraphs, the sample is the extension-compatible monthly panel, the model is the registered regime-interaction system, the estimator is pooled regime pricing and break diagnostics, the test assets are compatible anomaly portfolios, uncertainty is based on the registered Wald, Holm-adjusted, and break-test procedures, and economic magnitude is blocked until regime inputs exist.

10 Policy and Information Shocks

The high-frequency section is where identification language belongs. The selected shock-decomposition design distinguishes policy shock, central-bank-information, and ambiguous components before aggregating event-level observations to monthly factors. Outside this section, the AR residual remains a rate innovation.

For high-frequency shock paragraphs, the sample is the event-level Federal Reserve announcement panel aggregated to months after source terms are registered, the model is the policy, information, and ambiguous-component decomposition, the estimator is monthly aggregation plus spanning and pricing checks, the test assets are compatible anomaly portfolios, uncertainty follows the registered shock-decomposition diagnostics, and economic magnitude is blocked until event data are legally registered.

11 Out Of Sample Evaluation

The out-of-sample design is a falsification exercise. The initial training endpoint is 1999-12, and refits follow a predeclared annual schedule. Forecast records must store model vintage, training dates, factor definition, and asset universe. Null results and instability are not failures of the repository; they are outcomes the design is meant to preserve.

For out-of-sample paragraphs, the sample is the registered holdout window, the model is the confirmatory two-factor specification and benchmark set, the estimator is expanding and rolling no-lookahead forecasting, the test assets are compatible holdout anomaly portfolios, uncertainty is summarized by model-confidence-set and cross-sectional error diagnostics, and economic magnitude is blocked until forecast artifacts exist.

12 Interpretation

The intended interpretive categories are confirmation, narrowing, reinterpretation, generalisation, contradiction, and data-limited non-reproduction. The current evidence supports only the last category. That is a substantive statement about the repository state: the publication documents and software gates are in place, while the source-compatible generated panels are not. The appropriate conclusion is therefore bounded and procedural, not empirical.

The advantage of this conservative draft is that later results will have a clear place to land. If the baseline tables reproduce within tolerance, the extension can ask whether the same pricing structure survives later data and different monetary regimes. If the baseline can only be reconstructed approximately, the paper can still contribute by showing which inputs drive the difference. If the short-rate factor is weak, unstable, or absorbed by high-frequency components, that evidence will narrow the interpretation rather than being hidden.

13 Limitations

The main limitation is input availability. Exact public archive versions, author or WRDS portfolio files, source-compatible extension panels, high-frequency event files, and generated result cells are not yet available in the committed artifacts. These limitations are recorded in `research/data_access_matrix.csv`. The release gate reports no critical restricted-path finding, but it retains one major missing-input finding that blocks empirical-results release.

There are also econometric limitations that will remain relevant after data are registered. Cross-sectional pricing depends on informative beta dispersion. Regime samples may be short. Structural-break searches require multiplicity control. High-frequency event surprises require careful aggregation. Comparator models must use identical asset and date intersections. These are not optional robustness checks; they are conditions for interpreting the short-rate innovation as economically meaningful.

14 Reproducibility Statement

All empirical claims in this paper must be generated from versioned code, typed configuration, and machine-readable artifacts. The validation status is recorded in `VALIDATION.md`. Release artifacts include the release gate, sanitized environment manifest, software bill of materials, source checksums, source archive, archive manifest, and data-acquisition guide.

Table 2: Release Artifacts

Artifact	Machine-readable source
Release gate	<code>artifacts/release/release_gate.json</code>
Environment manifest	<code>artifacts/release/environment_manifest.json</code>
Software bill of materials	<code>artifacts/release/sbom.json</code>
Source checksums	<code>artifacts/release/source_artifact_checksums.sha256</code>
Source archive manifest	<code>artifacts/release/source_release_manifest.json</code>
Data acquisition guide	<code>docs/DATA_ACQUISITION.md</code>

15 Data Access Statement

Restricted article and supplement files remain local-only. The committed repository contains hashes and metadata but not copyrighted source documents, prompt files, licensed data extracts, local catalogs, or temporary artifacts. The data-acquisition guide explains which inputs must be obtained manually or verified before redistribution.

16 Hypothesis Registry

Confirmatory and extension hypotheses are listed in `research/hypothesis_registry.csv`. Hypotheses cannot be rewritten after observing generated errors, weak-factor diagnostics, regime instability, or forecast performance.

17 Robustness Appendix

The robustness appendix must be generated from `reports/generated/robustness_report.md` and registered robustness tables once empirical artifacts exist. Null results, weak-factor warnings, and instability are reportable outcomes.

18 Table-Level Replication Appendix

The table-level replication appendix must be generated from the registered replication audit artifact. No baseline verdict can include temporal, regime, shock-decomposition, or out-of-sample extension results.

19 Conclusion

This draft is a paper at the point where evidence discipline matters most. It does not claim that short-rate innovations price anomaly returns in the current repository state. It specifies how that claim will be tested, how replication and extension evidence will be separated, and how unavailable inputs will be reported. Once source-compatible empirical artifacts are generated, the same manuscript structure can carry numerical results without changing the rules used to judge them.

References

- Maio, P. F. and Santa-Clara, P. (2017). Short-term interest rates and stock market anomalies.
Journal of Financial and Quantitative Analysis, 52(3):927–961.